

# Identify and graph quadratic functions, solve quadratic equations graphically and numerically, and solve monic quadratic equations with integer roots algebraically, using graphing software and digital tools as appropriate

## Learning intention

- identify and graph quadratic functions, solve quadratic equations graphically and numerically, and solve monic quadratic equations with integer roots algebraically, using graphing software and digital tools as appropriate.

## Teach and model

- solving quadratic equations algebraically and comparing these to graphical solutions.
- recognising that the equation  $x^2=a$ , where  $a>0$ , has 2 solutions,  $x=\sqrt{a}$  and  $x=-\sqrt{a}$ .
- recognising that in a table of values, if the second difference between consecutive values of the dependent variable is constant.

## Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

## Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

## Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.